

Lec. 10

NULL SPACE

↓
matrix

Consider matrix A $\rightarrow m \times n$

$$\boxed{N(A)} = \{ \text{set of all vectors } \vec{x} \text{ such that } A\vec{x} = 0 \}$$

↑
subspace

$$\vec{x}_1, \vec{x}_2 \in N(A)$$

$$\Rightarrow \alpha \vec{x}_1 + \beta \vec{x}_2 \in N(A)$$

$$\vec{x}_1 \in N(A)$$

$$\Rightarrow A\vec{x}_1 = 0.$$

$$0 \in N(A) : A \cdot 0 = 0.$$

if there is any other vector $\vec{x}$

such that $\vec{x} \neq 0$, but $A\vec{x} = 0$,

then it is called a NON-TRIVIAL Nullspace.

For a square matrix A , if

Null space is non-trivial,

then A is NOT invertible.

$$\vec{x} \neq 0$$

But $A \bar{x} = 0$

if A^{-1} exists

$$A^{-1} A \bar{x} = A^{-1} 0 = 0$$

$$\Rightarrow \bar{x} = 0 \text{ (contradiction)}$$

$\Rightarrow A^{-1}$ does not exist

$\Rightarrow A$ NOT invertible.

Ex: Consider

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

Determine $N(A)$

Determine the BASIS of A .

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$$

Perform row operations

$$R_2: R_2 - R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix}$$

$$R_1: R_1 - R_2$$

2 pivots
 $\therefore \text{rank} = 2.$

$$\begin{bmatrix} \textcircled{1} & 0 & -1 & -2 & -3 \\ 0 & \textcircled{1} & 2 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 & -2 & -3 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$$

$$\Rightarrow \begin{aligned} x_1 - x_3 - 2x_4 - 3x_5 &= 0 \\ \Rightarrow x_1 &= x_3 + 2x_4 + 3x_5 \\ x_2 + 2x_3 + 3x_4 + 4x_5 &= 0. \end{aligned}$$

$$\Rightarrow x_2 = -2x_3 - 3x_4 - 4x_5$$

$$\vec{x} = N(A) = \begin{bmatrix} x_3 + 2x_4 + 3x_5 \\ -2x_3 - 3x_4 - 4x_5 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

no. of basis
 vectors

is 3

$\Rightarrow \text{Dim } N(A) = 3$

$$= x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 3 \\ -4 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

These vectors form a BASIS
 for $N(A)$

Since any $\vec{x} \in N(A)$
 can be expressed as a linear

combination of these 3 vectors.

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Dimension of $N(A)$ = NULLITY

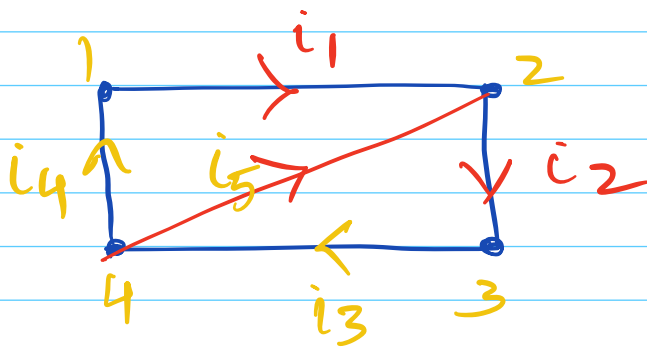
NULLITY of $A = 3$.

$$\text{Rank} + \text{Nullity} = 2 + 3 = 5$$

= # of columns
of A .

Applications of NULL space:

Consider circuit below:



Adjacency matrix: $A_{m \times n}$
of edges $\leftarrow$ m $\rightarrow$ # of nodes n

$$[A]_{ij} = \begin{cases} -1 & \text{if current } i \text{ leaves node } j \end{cases}$$

$\begin{cases} +1 & \text{if current } i \text{ enters node } j \\ 0 & \text{else.} \end{cases}$

$A = \begin{matrix} & \begin{matrix} \text{nodes} \\ 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} \text{currents} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \end{matrix}$

Adjacency matrix

Consider any current vector

$$\bar{i} = \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix}$$

$$\bar{i} \in N(A^T)$$

Adjacency matrix.

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = 0$$

$$\Rightarrow -i_4 + i_1 = 0 \Rightarrow i_4 = i_1$$

$\Rightarrow$ current entering node 1

is equal to current leaving node 1.

KIRCHHOFF'S current law

$$\underbrace{i_1 + i_5}_{\text{current entering node 2}} = \underbrace{i_2}_{\text{current leaving node 2}}$$

KCL for node 2.

$$\text{3rd row : } i_3 = i_4 + i_5$$

$$\text{4th row : } i_2 = i_3$$